

HAT-P-36b

R-1675

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_200320 · created 2026-08-02T10:58:35+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458928.6833 +/- 0.0034 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.155 +/- 0.022
Transit depth (Rp/Rs)^2	2.41 +/- 0.68 [%]
Orbital Inclination (inc)	80.21 +/- 2.82
Airmass coefficient 1 (a1)	0.966 +/- 0.029
Airmass coefficient 2 (a2)	0.016 +/- 0.011
Scatter in the residuals of the lightcurve fit is	3.01 %
Best Comparison Star	#3 - [550, 151]
Optimal Aperture	2.36
Optimal Annulus	10.24
Transit Duration (day)	0.072 +/- 0.021

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.155 ± 0.022 vs 0.126 (deviation 1.26σ)
Duration fit vs published	0.072 d vs 0.0925 d (deviation 0.93σ)
Mid-time O–C	6.6 min
Depth SNR	6.7
Residual scatter	3.01 %

Light curve

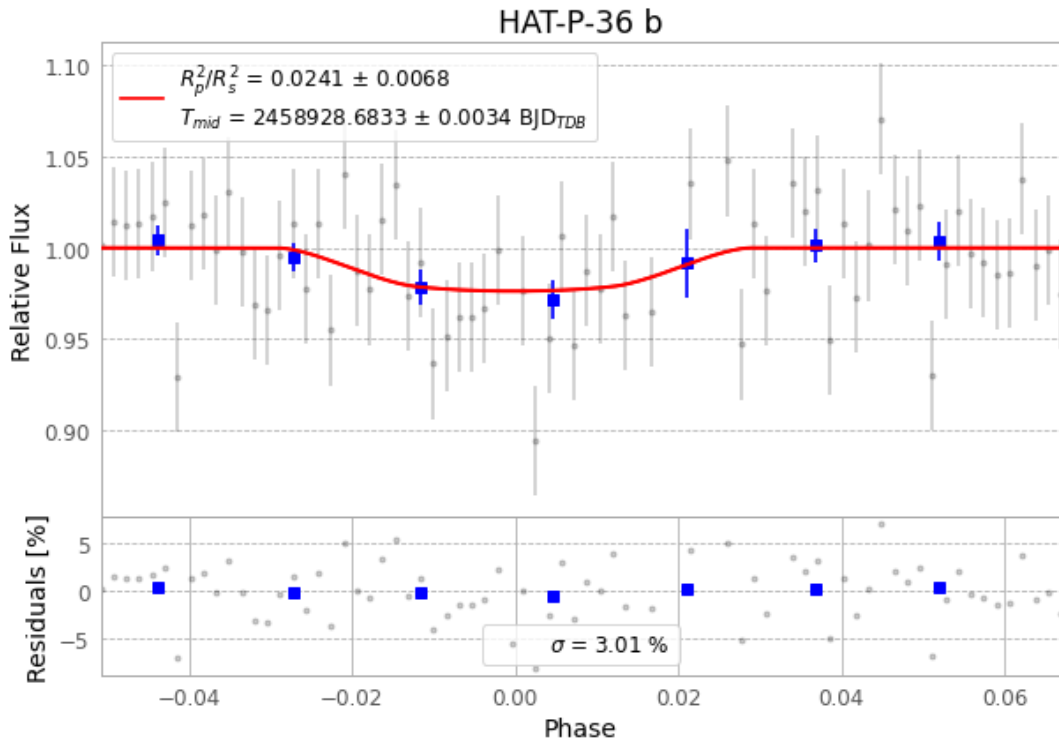


Figure 1 — FinalLightCurve_HAT-P-36 b_2020-03-19.png (SHA-256 6255c1ceafcf32fa...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [557, 167], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 31adb0f4fc563ebc...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

76 FITS frames (50 MB), HATP-36200320023912.FITS → HATP-36200320062412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-200320.log (b97ceccad2ee...), FinalLightCurve_HAT-P-36 b_2020-03-19.png (6255c1ceafcf...), FinalLightCurve_HAT-P-36 b_2020-03-19.csv (f9e02d6fe1cc...)

Compute resources

Wall time 165.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 10:58Z.